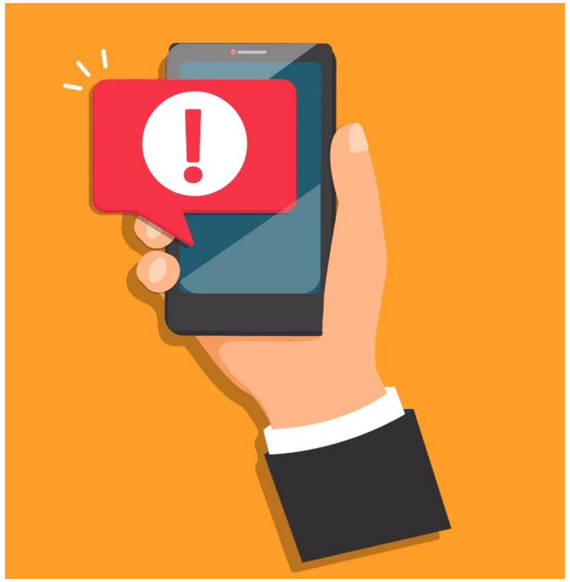
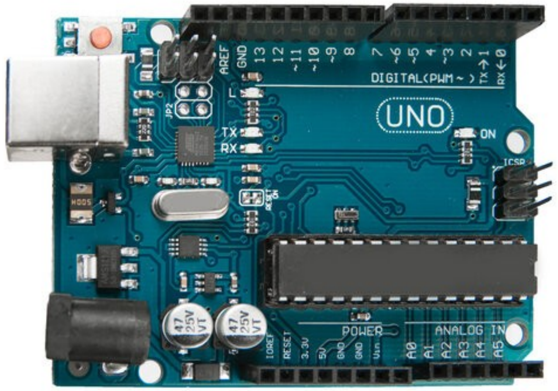
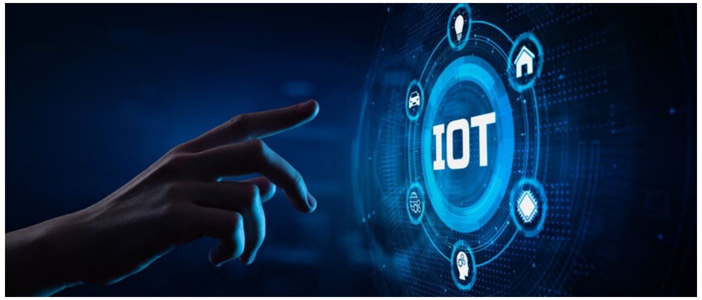






Cost-Effective IOT Framework for Enterprise Monitoring and Alerting

Spark Tank 2025 proposal

Overview

PROBLEM

SOLUTION

FUTURE CAPABILITIES

REQUESTED RESOURCES

CONCLUSION

Problem: Unable to detect and respond to events

Events

- Power loss
- Overheating
- Humidity/ flooding
- Research sensing (seismic, sound, machine signals)

Problem: Old technology cannot meet need

Old technology problems

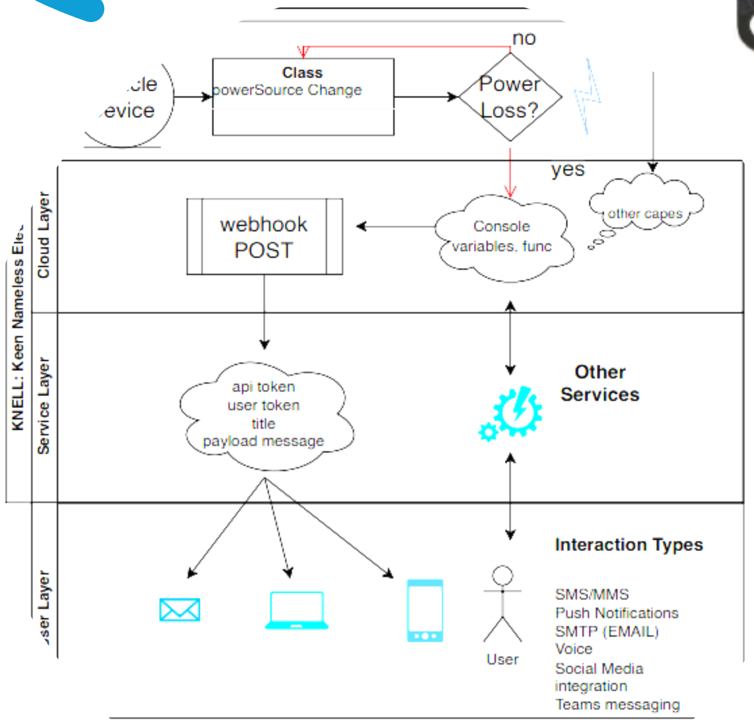
- Relies on grid power to operate
- Relies on ground networks to operate (power disruptions impact)
- Expensive with lengthy contract timelines
- Operated at base level: end units cannot customize
- Cannot accomplish remote sensing

Impact: Downtime, manpower, expensive equipment risk

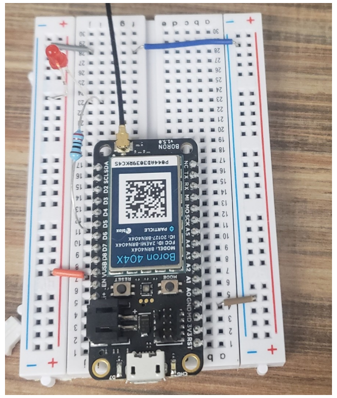
- Downtime: awaiting systems to be brought back up
- Manpower: monthly 24-Hour shifts to monitor buildings
- Equipment risk: expensive servers, physical security monitors, custom lab equipment

Solution: Device capable of **cellular** comm for monitoring

- Internet Of Things (IOT) devices are cheap
- Can leverage low-bandwidth cellular data paths off commercial network
- Customized programming & sensors
- Can integrate with cloud environments (AWS, Google cloud)



Process:



Solution: Generic, scalable, and easy to use

- Text alerts to users whenever power loss occurs
- New users added via mobile app, deploy new devices where needed
- Can update code over the air from cloud backend
- Custom sensors to meet user needs

Solution Impact: GOTS product enables future capabilities

SOLUTIONS CAN BE EXTENDED FOR HARDER PROBLEMS: GNSS, NTN

END POINT DATA COLLECTION FOR MACHINE LEARNING

END POINT AI / COMPUTER VISION

The Ask:

- 3-4 technical engineers assigned to help realize the product
- ~$2-4K for hardware design
- ~$1-2K for cloud resources

Conclusion

PROBLEM

SOLUTION

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CONCLUSIO N